

NEPHROLOGY TEACHING SERIES

Diabetic Kidney Disease

The four-pillar therapy

DKD is the #1 cause of end-stage kidney disease (ESKD). Every pillar in place at every visit.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why it matters

Most common cause of ESKD in the US — and highly modifiable if caught early.

WHAT THE VISIT COVERS

Confirm DKD (not another CKD cause)

Annual UACR + eGFR in all diabetics

4 pillars: ACEi/ARB, SGLT2i, finerenone, GLP-1 RA

A1c 7–8% individualized

SBP < 120 (AOBP)

Statin + lifestyle

Transplant planning by eGFR trend

TEACHING PEARL

Not all CKD in DM is DKD. Red flags: active sediment, rapid decline > 5 mL/min/yr, no retinopathy, short DM, abrupt nephrotic proteinuria. Biopsy when atypical.

Diagnosing DKD — pattern recognition

Triad: proteinuria + bland sediment + retinopathy + longstanding DM.

SUPPORTS DKD

- Long-standing diabetes (T1DM $\geq$ 10 yr; T2DM any duration)
- Gradual albuminuria (A2 $\rightarrow$ A3) followed by eGFR decline
- Bland UA (no RBC casts, minimal hematuria)
- Retinopathy present (esp. in T1DM)
- Other microvascular complications (neuropathy)
- Slow eGFR decline (2–5 mL/min/yr)

RED FLAGS FOR ALTERNATIVE DX

- Active sediment: RBC casts, dysmorphic RBCs
- Rapid decline $>$ 5 mL/min/yr
- No retinopathy in T1DM with proteinuria
- Proteinuria appears $<$ 5 yr after T1DM diagnosis
- Abrupt nephrotic-range proteinuria
- Systemic features: rash, arthritis, hemoptysis

Consider biopsy when these are present

The four pillars

Sequential, but aim to have all four in place when appropriate.

1. ACEi / ARB

First-line for albuminuria

Titrate to max dose

Accept Cr $\uparrow$ up to 30%

Hold if K $>$ 5.5

2. SGLT2 INHIBITOR

Dapa or empa 10 mg daily

Start at eGFR $\geq$ 20

Expect early eGFR dip — don't stop

DKA watch (T1DM, sick days)

3. FINERENONE

T2DM + CKD; K $\leq$ 5.0 at start

10 mg if eGFR 25 to $<$ 60

20 mg if eGFR $\geq$ 60

Additive to ACEi/ARB + SGLT2i

4. GLP-1 RA

Semaglutide 1.0 mg SQ weekly

Kidney + CV benefit (FLOW 2024)

Weight loss + glycemic bonus

GI AEs — titrate slowly

Glycemic control in CKD

Individualized targets. Hypoglycemia risk rises as eGFR falls.

Scenario	A1c target	Notes
Young, short DM, low hypoglycemia risk	< 7%	Tight control — early preserves nephrons
Established DKD, moderate comorbid	7.0–7.5%	Balance benefit vs hypoglycemia risk
Advanced CKD (eGFR < 30), elderly, hx hypoglycemia	7.5–8%	Watch insulin accumulation; prefer short-acting
Dialysis, high hypoglycemia risk	8–8.5%	A1c unreliable (anemia, transfusion); use continuous glucose monitor (CGM)

Drugs that need adjustment as eGFR falls:

- Metformin: stop at eGFR < 30; dose-reduce 30–45
- Sulfonylureas: glipizide preferred (no renal clearance); avoid glyburide
- Insulin: doses often need reduction by 25–50% as eGFR < 30
- dipeptidyl peptidase-4 (DPP-4): OK but dose adjust (except linagliptin)
- SGLT2i: start at ≥ 20 , continue to KRT
- GLP-1 RA: OK across eGFR range
- Pioglitazone: avoid if HF risk.

BP, lipids, and CV risk

DKD patients die from CV disease more often than ESKD.

BLOOD PRESSURE

Target SBP < 120 by AOBP (KDIGO 2021)

Home BP < 125/75 average (lower than office)

First-line: ACEi/ARB (dual purpose)

Add: thiazide-like if eGFR $\geq$ 30; loop if < 30; DHP CCB as needed

Resistant: add spironolactone or finerenone

**Avoid dual renin-angiotensin-aldosterone system (RAAS)
(ONTARGET — harm)**

Salt restriction < 2 g Na/day

LIPIDS + CV

Diabetes age 40–75: at least moderate-intensity statin

CKD and high ASCVD risk → high-intensity or add-on therapy when indicated

SHARP: statin + ezetimibe ↓ atherosclerotic events in CKD (not dialysis-specific benefit)

Lipid panel annually; LDL < 70 aggressive in ASCVD

Smoking cessation > any drug for CV reduction

Aspirin only if established ASCVD (not primary prevention in CKD)

Vaccinate: flu, COVID, pneumococcal (PCV20 alone, OR PCV15 → PPSV23 — ACIP 2024), HBV

Monitoring and escalation

Visit cadence, labs, and when to refer.

Stage / risk	Visit cadence	Labs
Low risk (G1-2, A1)	Annual	BMP, UACR, A1c, lipid
Moderate (G1-2, A2 or G3a, A1)	Every 6 mo	+ CBC, bicarb, PO4, PTH if stage 3
High (G3a/A2, G3b/A1)	Every 3-4 mo	+ Hgb, ferritin / TSAT, vit D, Ca
Very high (G3b+/A2+ or G4)	Every 3 mo or less	All of above; kidney failure risk (KFRE)
G4-5	Monthly if accelerating	Modality education, transplant referral, access planning

Landmark DKD trials

The four-pillar story in chronological order.

Trial	Year	Takeaway
Captopril Trial	1993	T1DM: captopril ↓ doubling Cr 48%. Foundation of RAAS in DKD.
RENAAL	2001	T2DM: losartan ↓ ESKD/doubling Cr 25%.
IDNT	2001	T2DM: irbesartan ↓ progression 20–23%.
ONTARGET	2008	Dual RAAS harmful — never combine ACEi + ARB.
CREDENCE	2019	Canagliflozin ↓ ESKD/doubling Cr/renal-CV death 30% in T2DM + CKD.
DAPA-CKD	2020	Dapagliflozin ↓ kidney failure 39% across DM + non-DM CKD.
EMPA-KIDNEY	2023	Empagliflozin ↓ CKD progression 28% across eGFR 20–90.
FIDELIO-DKD	2020	Finerenone ↓ CKD progression 18% on max RAAS.
FIGARO-DKD	2021	Finerenone ↓ CV events 13%, HF hosp 29%.
FLOW	2024	Semaglutide ↓ kidney disease progression 24% in T2DM + CKD.

Full citations on references slide.

Common DKD clinic mistakes

The pillars left out, the drugs given wrong.

AVOID

- Skipping annual UACR
- Stopping RAAS for 20% Cr rise
- Withholding SGLT2i at low eGFR
- Metformin at eGFR < 30
- Dual RAAS blockade
- No retinopathy screen
- Late nephrology referral

DO INSTEAD

- Annual UACR + eGFR together
- Accept Cr ↑ up to 30% on RAAS
- SGLT2i at eGFR ≥ 20; continue to KRT
- Stop metformin < 30; halve 30–45
- Max-titrate one RAAS
- Retina referral yearly
- Refer neph at G3b–A3 or G4

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 45-year-old woman with T2DM × 12 yr, proliferative retinopathy (post-laser), A1c 8.4%, BP 142/88 home avg. Meds: metformin 1 g BID, glipizide 10 mg, lisinopril 20 mg, HCTZ 12.5 mg. Labs: eGFR 42, UACR 580, K⁺ 4.5, Hb 11.2, LDL 118 (on no statin).

QUESTION: Audit her DKD care — what are the top 4 things missing?

Clinical case — answer

Key teaching points from the case.

ANSWER

(1) Lisinopril at SUB-maximal dose (should be 40), (2) No SGLT2i (start dapagliflozin 10 mg — eGFR qualifies), (3) No finerenone (eligible: K⁺ 4.5, eGFR 25-60 → start 10 mg), (4) No statin (start moderate-to-high intensity).

TEACHING

Classic DKD presentation with retinopathy — diagnosis secure. All 4 pillars should be ON when eligible: ACEi/ARB max dose, SGLT2i, finerenone, GLP-1 RA. The retinopathy supports microvascular disease, but keep an eye out for atypical features. Titrate lisinopril to 40 (accept Cr rise to 30%). Start SGLT2i at eGFR $\geq$ 20. Finerenone 10 mg with K⁺ recheck in 1 month. GLP-1 RA (semaglutide 1 mg weekly per FLOW) adds kidney and CV benefit, also helps A1c and weight. Statin: diabetes age 40-75 is already an indication; CKD pushes intensity higher. BP target SBP < 120 when measured by standardized AOBP and tolerated. A1c 7-8% range given CKD.

REFERENCES

Diabetic Kidney Disease

Guidelines

KDIGO 2022 Diabetes in CKD
KDIGO 2024 CKD Guideline
KDIGO 2021 BP in CKD
ADA 2026 Standards of Care
ACC/AHA guidelines for CV in CKD

Landmark Trials

Captopril Trial — NEJM 1993
RENAAL — NEJM 2001
IDNT — NEJM 2001
ONTARGET — NEJM 2008
CREDENCE — NEJM 2019
DAPA-CKD — NEJM 2020
EMPA-KIDNEY — NEJM 2023
FIDELIO-DKD — NEJM 2020
FIGARO-DKD — NEJM 2021
FLOW — NEJM 2024

UpToDate (2026 access):

Diabetic kidney disease: Manifestations, evaluation, and diagnosis • Treatment of diabetic kidney disease • SGLT2 inhibitors in CKD • Mineralocorticoid receptor antagonists in CKD • GLP-1 receptor agonists in CKD